DISPROOF OF EVENTUAL EQUALITY IN ERDŐS PROBLEM 878 (QUESTION 3B)

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For an integer $n \geq 1$, let $\mathcal{P}(n)$ denote the set of prime factors of n, and for a prime p let $q_p(n) := p^{\lfloor \log_p n \rfloor}$ be the largest power of p not exceeding n. Erdős [1] defined

$$f(n) := \sum_{p \in \mathcal{P}(n)} q_p(n), \qquad F(n) := \max \sum_{a \in A} a,$$

where $A \subseteq [2, n]$ ranges over finite sets of pairwise coprime integers a with $\mathcal{P}(a) \subseteq \mathcal{P}(n)$, and asked whether the maxima $\mathrm{maxUpTo}\, f\, x := \max_{n \leq x} f(n)$ and $\mathrm{maxUpTo}\, F\, x := \max_{n \leq x} F(n)$ satisfy $\mathrm{maxUpTo}\, f\, x = \mathrm{maxUpTo}\, F\, x$ for all sufficiently large x.

Theorem. *For every $X_0 \in \mathbb{N}$, there exists $x \geq X_0$ such that* $\mathrm{maxUpTo}\, f\, x < \mathrm{maxUpTo}\, F\, x$.

In particular, $\mathrm{maxUpTo}\, f\, x \neq \mathrm{maxUpTo}\, F\, x$ for arbitrarily large x, so the answer to Question 3b is negative.

Correspondence with the Lean formalization. The proof below is the one formalized in `FormalConjectures/ErdosProblems/878.lean`, step by step and with the same names ($q_1, \dots, q_6$, Q_6, C_6, N_{smooth}, N_2, A, M, step, rem, K, Y_0, Y_a, Y_b, Q_a, Q_b, R', D, r_a, r_b, b_i, J, ℓ, z_j, ...). All variables are natural numbers, except $\mathrm{rem}(k)$, which is real. Some inequalities are written in the opposite direction, or with fractions where Lean clears denominators (e.g. $Q_a \geq Y_a/14$ for $Y_a \leq 14Q_a$, and $\sum_{q \in Q_6} q_q(n) < \frac{6}{7}n$ for $7\sum_{q \in Q_6} q_q(n) < 6n$), and a few are stated in a stronger form than Lean needs (e.g. strict instead of non-strict). The remaining correspondences are:

This note	Lean
$\mathcal{P}(n)$; $\lfloor \log_p n \rfloor$; $q_p(n)$	`n.primeFactors`; `Nat.log p n`; `q_p p n` (in Lemma 2, $q_2(\cdot)$ is `q_p 2`, not the constant `q2`)
$\lfloor a/b \rfloor$; $[a, b]$; $(a, b]$	`a / b` in $\mathbb{N}$; `Finset.Icc a b`; $a < \cdot \leq b$
$\mathrm{maxUpTo}\, f\, x$	`maxUpTo f x` (maximum over $n \in \{0, \dots, x\}$; as $f(0) = F(0) = 0$ in Lean, including $n = 0$ does not change the maximum)
an integer $a \in [2, n]$ with $\mathcal{P}(a) \subseteq P$	`IsAdmissibleTerm n P a`
$14 q_p(\lfloor 21Y/20 \rfloor) < Y$	`IsGoodForPrime p Y`
$q_1, \dots, q_6$ are six distinct primes; $C_6 > 0$	`Q6_prime`, `Q6_card`; `C6_pos`
Theorem; the sentence after it	`erdos_878_q3_disproved`; `erdos_878.parts.iii_eventually`
Lemma 1	`lemma1_exists_smooth_gt`; the choice of j is `exists_step_crossing`
inequality (1)	`exists_F_ge_of_smooth`, via `q_p_admissible` and `sum_le_F_of_disjoint`
Lemma 2	`lemma2_f_two_mul_gt`
Lemma 3	`lemma3_superlinear_maxUpTo_f`
Lemma 4	`lemma4_exists_good_Y`
contradiction hypothesis	`h_no_jump`
Step 1	`single_step_jump`; the decomposition of $f(n)$ is `sum_sdiff_erase_two_le`, and $r_a r_b \prod_{p \in R'} p \leq n$ is `mul_mul_prod_erase_le`
Step 2: iterating Step 1; rem; $20(z + \mathrm{rem}(k)) \leq 21Y \Rightarrow z \leq \lfloor 21Y/20 \rfloor$; the induction on k	`iter`; `rem`; `le_cap`; `h_seq`

This note	Lean
monotonicity of q_p and maxUpTo; $g(n) \le \mathrm{maxUpTo}\, g\, x$ for $n \le x$ $(g = f, F)$	`q_p_mono`, `maxUpTo_mono`; `le_maxUpTo`

The Lean file states Lemma 2 before Lemma 1, and Step 1 as the separate lemma `single_step_jump` before Lemmas 3 and 4; its hypotheses beyond those of Step 1 (inequality (1), the contradiction hypothesis for $x \ge Y$, $20 \le Y$, $2N_{\mathrm{smooth}} \le Y$, and the conclusion of Lemma 4 for Y) are the standing assumptions of Step 1 here.

Lemma 1. *If $p < r$ are primes with $98r < 100p$, then for every $n \ge p^{p^2+2}$ there exists an integer $z \in [2, n]$ with $\mathcal{P}(z) \subseteq \{p, r\}$ and $98n < 100z$.*

Proof. Put $\ell = \lfloor \log_p n \rfloor$, $J = p^2 + 1 < \ell$, and $z_j = p^{\ell-j} r^j$. Then $z_0 = q_p(n) \le n$, while

$$z_J \ge p^{\ell-J}(p+1)^J \ge p^{\ell-J}(p^J + Jp^{J-1}) = p^{\ell-J}\big(p^J + (p^2+1)p^{J-1}\big) > p^{\ell+1} > n.$$

Choose $0 \le j < J$ with $z_j \le n < z_{j+1}$. Since $z_{j+1}p = z_j r$, we have $98np < 98z_{j+1}p = 98z_j r < 100z_j p$, so $z = z_j$ has the required properties. □

Fix the six primes

$$q_1 := 168000061, \quad q_2 := 168000101, \quad q_3 := 168000121,$$
$$q_4 := 168000139, \quad q_5 := 168000143, \quad q_6 := 168000167,$$

let $Q_6 := \{q_1, \dots, q_6\}$, and let $C_6 := \prod_{p \in Q_6} p$. Applying Lemma 1 to the pairs (q_1, q_2), (q_3, q_4), and (q_5, q_6) (each satisfies $q_{2i-1} < q_{2i}$ and $98q_{2i} < 100q_{2i-1}$) yields the threshold $N_{\mathrm{smooth}} := q_1^{q_1^2+2} + q_3^{q_3^2+2} + q_5^{q_5^2+2}$, which exceeds each $q_{2i-1}^{q_{2i-1}^2+2}$, such that for every $m \ge N_{\mathrm{smooth}}$ there exist $b_1, b_2, b_3 \in [2, m]$ with $\mathcal{P}(b_i) \subseteq \{q_{2i-1}, q_{2i}\}$ and $100b_i > 98m$. Whenever $m \ge N_{\mathrm{smooth}}$, $R' \subseteq \mathcal{P}(m)$ with $R' \cap Q_6 = \emptyset$, and $Q_6 \subseteq \mathcal{P}(m)$, the integers b_1, b_2, b_3 and $q_p(m) \in [2, m]$ $(p \in R')$ are pairwise coprime with prime factors dividing m, so

$$100F(m) > 294m + 100 \sum_{p \in R'} q_p(m). \tag{1}$$

Lemma 2. *For all $n \ge 1$, $f(n) < f(2n)$.*

Proof. Passing from n to $2n$ never decreases q_p for $p \in \mathcal{P}(n)$, and strictly increases the $p = 2$ term (either $q_2(2n) = 2q_2(n)$ if $2 \mid n$, or $q_2(2n) \ge 2$ is adjoined if $2 \nmid n$). □

Lemma 3. *For every $R \in \mathbb{N}$, there is N_2 such that $\mathrm{maxUpTo}\, f\, y > Ry$ for all $y \ge N_2$.*

Proof. Since $\sum_p 1/p = \infty$, choose N with $\sum_{p<N} 1/p \ge 2R + 1$ and set $P = \prod_{p<N} p$ and $N_2 = 2P$. For $y \ge N_2$, let $m = P\lfloor y/P \rfloor > y/2$. Every prime $p < N$ divides m and satisfies $q_p(m) > m/p$, so

$$\mathrm{maxUpTo}\, f\, y \ge f(m) \ge \sum_{p<N} q_p(m) \ge m \sum_{p<N} \frac{1}{p} \ge (2R+1)m > Ry. \qquad \square$$

Lemma 4. *For every X_0, there is $Y \ge \max(X_0, 20)$ with $14q_p(\lfloor 21Y/20 \rfloor) < Y$ for all $p \in Q_6$.*

Proof. Let $Y_0 = \max(X_0, 20)$. If a prime $p \in Q_6$ fails the inequality for $Y_a := Y_0 \cdot 15^{m_1}$ and $Y_b := Y_0 \cdot 15^{m_2}$ with $0 \le m_1 < m_2 \le 6$, then $Q_a := q_p(\lfloor 21Y_a/20 \rfloor) \ge Y_a/14$ and $Q_b := q_p(\lfloor 21Y_b/20 \rfloor) \ge Y_b/14 \ge \frac{15}{14} Y_a > \frac{21}{20} Y_a \ge Q_a$. Since Q_a, Q_b are powers of p,

$$\frac{21}{20} Y_b \ge Q_b \ge pQ_a \ge \frac{168000000}{14} Y_a = 12000000 Y_a,$$

contradicting $Y_b \le 15^6 Y_a = 11390625 Y_a$ (note that $\frac{21}{20} \cdot 11390625 = 11960156.25 < 12000000$). Thus each of the 6 primes in Q_6 eliminates at most one of the 7 exponents $m \in \{0, \dots, 6\}$, and some $Y = Y_0 \cdot 15^m$ satisfies the lemma. □

Proof of the Theorem. Suppose for contradiction that $\operatorname{maxUpTo} F\, x \le \operatorname{maxUpTo} f\, x$ for all $x \ge X_0$. Put $A := 2000C_6{+}1$. By Lemma 3 with $R = 2(A{+}10)$, there is N_2 such that $\operatorname{maxUpTo} f\, y > 2(A+10)y$ for all $y \ge N_2$. Let Y be given by Lemma 4 applied to $\max(X_0, N_2, 2N_{\mathrm{smooth}})$, so that $Y \ge \max(X_0, N_2, 20, 2N_{\mathrm{smooth}})$ and $14q_p(\lfloor 21Y/20\rfloor) < Y$ for all $p \in Q_6$.

Step 1. Suppose $y \in [Y, \lfloor 21Y/20\rfloor]$ and $M \ge 1$ satisfy $100 \operatorname{maxUpTo} f\, y \ge 105(M+8)Y$. Choose $n \le y$ with $f(n) = \operatorname{maxUpTo} f\, y$.

- Then $n \ge 1$, as $f(n) = \operatorname{maxUpTo} f\, y \ge \frac{105(M+8)Y}{100} > 0$. If $2n \le y$, then $f(2n) \le \operatorname{maxUpTo} f\, y = f(n)$, contradicting Lemma 2; hence $y < 2n$, and $n > y/2 \ge N_{\mathrm{smooth}}$.
- Since $f(n) \le |\mathcal{P}(n)|\, n$ and $20n \le 21Y$, we have $105(M+8)Y \le 100|\mathcal{P}(n)|\, n \le 105|\mathcal{P}(n)|\, Y$, so $|\mathcal{P}(n)| \ge M+8$.
- Removing Q_6 (≤ 6 primes) and primes $< M$ ($\le M$ integers) from $\mathcal{P}(n)$ leaves at least two distinct primes $r_a, r_b \ge M$ outside Q_6.
- Put $R' := (\mathcal{P}(n) \setminus Q_6) \setminus \{r_a, r_b\}$ and $D := \big(\prod_{p\in R'} p\big)C_6$. Since $r_a r_b \prod_{p\in R'} p$ divides n, it is at most $n \le 2Y$; as $M^2 \le r_a r_b$, we get $M^2 D \le 2C_6 Y$, that is,
$$D \le \left\lfloor \frac{2C_6 Y}{M^2} \right\rfloor.$$
- Let $m := D(\lfloor n/D\rfloor + 1) \in (n, y+D]$. By Lemma 4, each $q \in Q_6$ satisfies $14q_q(n) \le 14q_q(\lfloor 21Y/20\rfloor) < Y < 2n$, so $\sum_{q\in Q_6} q_q(n) < \frac{6}{7}n$. Since every prime factor of n lies in $\{r_a, r_b\} \cup R' \cup Q_6$, we have $f(n) \le q_{r_a}(n) + q_{r_b}(n) + \sum_{p\in R'} q_p(n) + \sum_{q\in Q_6} q_q(n)$. Using $q_{r_a}(n), q_{r_b}(n) \le n$ and $q_p(n) \le q_p(m)$ for $p \in R'$,
$$f(n) < 2n + \frac{6}{7}n + \sum_{p\in R'} q_p(n) \le \frac{20}{7}n + \sum_{p\in R'} q_p(m).$$
Since $D \mid m$, we have $R' \subseteq \mathcal{P}(m)$ and $Q_6 \subseteq \mathcal{P}(m)$; also $R' \cap Q_6 = \emptyset$ and $m > n \ge N_{\mathrm{smooth}}$. So comparing with (1) gives $F(m) - f(n) > \big(\frac{294}{100} - \frac{20}{7}\big)n = \frac{29}{350}n > \frac{29}{700}y \ge \frac{29}{700}Y > \frac{Y}{25}$. Since $m \le y + D$ and $\operatorname{maxUpTo} F\,(y+D) \le \operatorname{maxUpTo} f\,(y+D)$ (as $y + D \ge Y \ge X_0$), setting $y' = y + D$ yields

$$y \le y' \le y + \left\lfloor \frac{2C_6 Y}{M^2} \right\rfloor \quad \text{and} \quad 25 \operatorname{maxUpTo} f\, y' \ge 25 \operatorname{maxUpTo} f\, y + Y. \tag{2}$$

Step 2. Iterating Step 1 gives: if $Y \le y$, $M \ge 1$, $100 \operatorname{maxUpTo} f\, y \ge 105(M+8)Y$, and $y + j\lfloor 2C_6 Y/M^2\rfloor \le \lfloor 21Y/20\rfloor$, then there is $y' \ge y$ with $y' \le y + j\lfloor 2C_6 Y/M^2\rfloor$ and $25 \operatorname{maxUpTo} f\, y' \ge 25 \operatorname{maxUpTo} f\, y + jY$. This follows by induction on j (for $j = 0$ take $y' = y$), since the y' obtained for j satisfies $Y \le y' \le \lfloor 21Y/20\rfloor$ and $\operatorname{maxUpTo} f\, y' \ge \operatorname{maxUpTo} f\, y$, so that Step 1 applies to it.

Let $\operatorname{rem}(k) := \frac{54C_6 Y}{A+k}$, and note that $20(z + \operatorname{rem}(k)) \le 21Y$ implies $z \le \lfloor 21Y/20\rfloor$, since $\operatorname{rem}(k) \ge 0$ and $z \in \mathbb{N}$. We construct $y_k \ge Y$ by induction on $k \ge 0$ with

$$20(y_k + \operatorname{rem}(k)) \le 21Y \quad \text{and} \quad 100 \operatorname{maxUpTo} f\, y_k \ge 105(A + 1 + k + 8)Y.$$

For $k = 0$, $y_0 = Y$ satisfies both inequalities since $20 \operatorname{rem}(0) = \frac{1080C_6 Y}{2000C_6+1} < Y$ and $100 \operatorname{maxUpTo} f\, Y > 200(A+10)Y \ge 105(A+1+0+8)Y$. Given y_k, set $M = A + 1 + k \ge 1$ and $\mathrm{step} = \lfloor 2C_6 Y/M^2\rfloor$. Since $\mathrm{step} \cdot M^2 \le 2C_6 Y$ and $(A+k)(A+k+1) \le M^2$,

$$27 \cdot \mathrm{step} \le \frac{54C_6 Y}{(A+k)(A+k+1)} = \operatorname{rem}(k) - \operatorname{rem}(k+1),$$

so $20(y_k + 27\cdot\mathrm{step} + \operatorname{rem}(k+1)) \le 20(y_k + \operatorname{rem}(k)) \le 21Y$ and hence $y_k + 27\cdot\mathrm{step} \le \lfloor 21Y/20\rfloor$. Iterating Step 1 27 times from y_k (with this M, for which $105(M+8)Y = 105(A+1+k+8)Y$) therefore gives y_{k+1} with $y_k \le y_{k+1} \le y_k + 27\cdot\mathrm{step}$, so that $20(y_{k+1} + \operatorname{rem}(k+1)) \le 21Y$, and $100 \operatorname{maxUpTo} f\, y_{k+1} \ge 100 \operatorname{maxUpTo} f\, y_k + 27 \cdot 4Y \ge 105(A+1+k+8)Y + 108Y > 105(A+1+(k+1)+8)Y$.

Taking $K = \operatorname{maxUpTo} f \lfloor 21Y/20 \rfloor + 1$, the first inequality for y_K gives $y_K \leq \lfloor 21Y/20 \rfloor$, hence $\operatorname{maxUpTo} f\, y_K \leq \operatorname{maxUpTo} f \lfloor 21Y/20 \rfloor < K$, while the second gives $100 \operatorname{maxUpTo} f\, y_K \geq 105(A+1+K+8)Y > 100K$, a contradiction. □

References

[1] P. Erdős, *On two unconventional number theoretic functions and on some related problems*, Topics in classical number theory, Vol. I, II (Budapest, 1981), Colloq. Math. Soc. János Bolyai **34** (1984), 445–458.